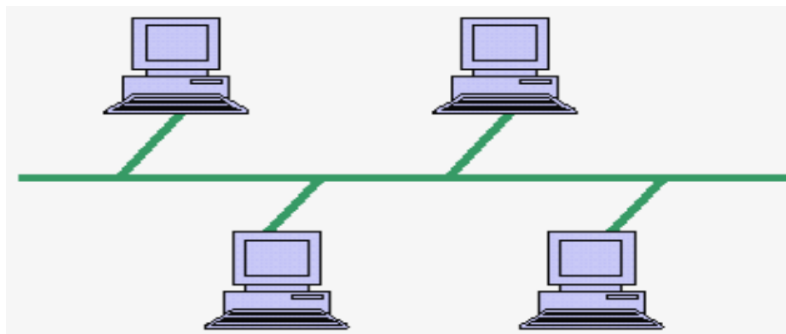


Network Topologies

Network Topology could be broadly classified into following different types. These different types are as follows:

- Bus Topology
- Mesh Topology
- Ring Topology
- Star Topology
- Tree Topology
- Point-to-Multipoint Topology
- Point-to-Point Topology

1. Bus Topology



1. A bus topology is a type of network in which each node present for communication is connected to a wire.
2. This wire is responsible for transmitting a message from one participant node to another receiver node
3. Generally, this wire is termed or known as “Backbone”
4. The connecting wire is rightly termed as Backbone since the completely smooth working of the network depends upon backbone wire
5. It is sometimes also referred to as Line Topology

Uses:

1. It is generally used over a small network such as LAN

2. Here messages need to be transmitted over the limited number of participant nodes

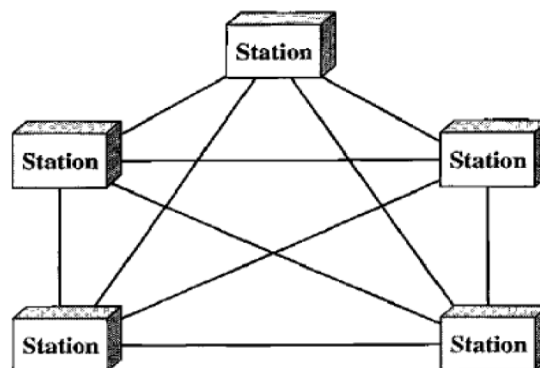
Advantages:

1. Easy to use and understand
2. Cost-effective
3. Easy to extend the network by simply adding repeater (a repeater generally boosts the signal and helps for transmission to longer distance)

Disadvantages:

1. A single cable break will bring down the bus topology
2. Too many participants nodes make the network slow

2. Mesh Topology



1. As the name suggests, Mesh Topology generally forms a mesh of all the interconnected nodes
2. Here each node is connected to every other node, through a single one-to-one communicating wire
3. There is a one-to-one mapping between each node

Uses:

1. A mesh topology is generally used over a large network where a breakdown of single node shall not affect the complete network
2. It is generally used as if a single fault in routes, wires, etc shall not hamper complete topology

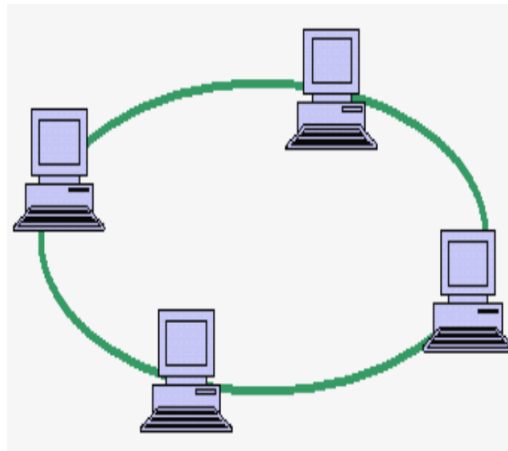
Advantages:

1. Better fault-tolerance capacity
2. Failure of one participant nodes or other devices such as routers will not affect the complete network
3. If there is a breakdown in one path between two nodes then alternate path is always available

Disadvantages:

1. The network becomes too complex as there are large participants nodes
2. Becomes costly due to the set-up of multiple paths

3. Ring Topology



1. As the name suggests ring topology forms a ring by connecting participants nodes
2. Nodes are connected in such a way that a single wire forms the path whose ends are joined to form a circle

3. Ring topology uses the concept of the token where the token is transmitted along with the message for correct delivery of the message

Uses:

1. A ring topology is generally designed in such a way that it forms a ring type network either physically or logically
2. These rings are made around campus or building to form a high speed flexible, reliable network

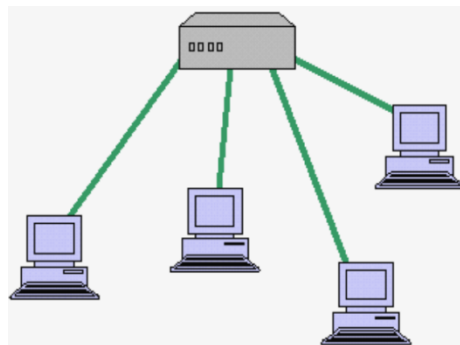
Advantages:

1. No master-slave concept
2. Each node has its own share of responsibility
3. Could work in high capacity network

Disadvantages:

1. A single node failure affects complete network
2. Sometimes it becomes difficult to troubleshoot if the network is too large
3. Adding or manipulating the network affects other participants nodes

4. Star Topology



1. A star topology generally consists of a central hub
2. Every participant node is directly connected to this hub
3. Hub acts as a central point to receive the message from the sender node and then transmits to other participant nodes
4. There is no direct connection between nodes

Uses:

1. A general implementation of the star topology is an Ethernet 10BaseT network

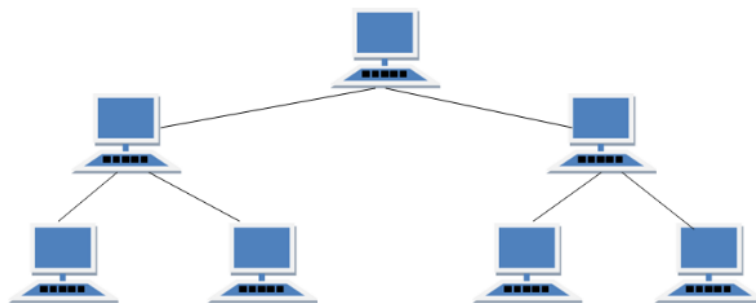
Advantages:

1. A single node failure does not affect the complete network
2. The network can run smoothly as far as the centralized hub is running smoothly
3. Is more cost-effective since the centralized network reduces network administration cost highly

Disadvantages:

1. Failure of the central hub will disrupt the whole topology
2. It is slightly costly when compared to bus topology

5. Tree Topology



1. A tree topology is a special type of network where many participants nodes are connected to form a tree type of architecture

2. Generally, there is a central initiator node which could be compared as the root of a tree.
3. This root node on extending further has different nodes connected to it which could be compared as branches of a tree
4. A tree topology is a combination of bus topology and star topology

Uses:

1. It is generally used where there is a requirement of communication between two networks

Advantages:

1. Easily scalable since adding new node a leaf node is easily possible
2. Leaf nodes could accommodate new nodes thus forming the hierarchical chain
3. Other hierarchical nodes do not get affected if particular nodes fail
4. Easy to debug

Disadvantages:

1. A huge length of wires are required to form the hierarchical chain
2. A lot of maintenance is required

6. Point-to-Multipoint Topology

1. A Point-to-Multipoint Topology consists of one node which acts as a sender and on the other side are multiple nodes which act as receiver

Uses:

1. A radio channel broadcast or television is a typical example of it

Advantages:

1. Fast
2. Easy to widespread a message when needs to broadcast to a large number of nodes

Disadvantages:

1. Expensive
2. Initial set-up is quite high

7. Point-to-Point Topology

1. A Point-to-Point Topology consists of two nodes which are communicating directly. A typical example is a telephonic call where two nodes are connected for communication.

Uses:

1. A telephonic call, two way radio communicating is a typical example of it

Advantages:

1. Fast and secure
2. Single node failure does not affect other

Disadvantages:

1. Expensive
2. High maintenance cost